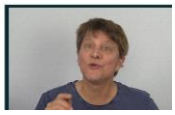
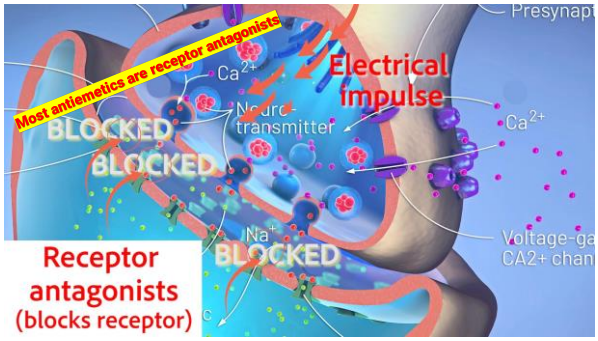

YOU GOT THIS!



1


Neurons, neurotransmitters, emesis and antiemetic drugs

2


Most antiemetics are receptor antagonists
Electrical impulse
BLOCKED
BLOCKED
BLOCKED
Receptor antagonists (blocks receptor)

3

Safety issues associated with vomiting

* Electrolyte abnormalities
* Dehydration
* Acid-base abnormalities

Any of these can in turn cause neurological or cardiac emergencies

4

Determine the underlying cause

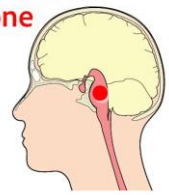
* Intestinal obstruction
* Appendicitis
* Gallstones
* Perforated bowel

5

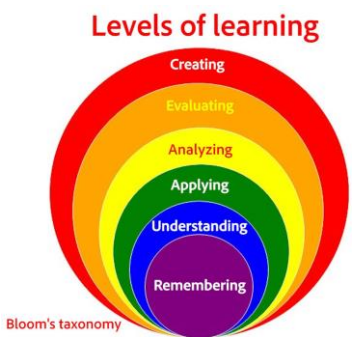
Imbalances can cause emesis

* Glucose
* Electrolytes
* Toxins
* Acid/base
* Thyroid
* Many DRUGS

Chemoreceptor trigger zone (CRTZ)



6





7

Diabetes mellitus type 1

* High levels of glucose
  Detected in urine and blood
* High levels of ketones
  Sweet smelling breath and urine
  Detected in urine and blood

8



Vomiting centre

*Uses H1 and ACh

*Activated by 4 paths

Path 1: Cerebral cortex
Example: extreme pain
Drug that addresses nausea
from this area: Benzodiazepines

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Vomiting centre

*Uses H1 and ACh

*Activated by 4 paths

Path 4: Vestibular area (uses H1 and ACh)

Example: motion sickness

Drugs that address nausea

H1 antagonists and the Ach antagonists

The vestibular nerve transmits impulses from the inner ear

10

The next few slides picture the drugs good for motion sickness

11



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Medications for motion sickness

This pathway uses the neurotransmitters:

Histamine 1 and AcetylCHOLamine

AntiHISTAMINEs AntiCHOLinergics

EXAMPLE:
Hyoscine
(Scopolamine)

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Anticholinergic agents

(against acethycholine)

Hyoscine (scopolamine)

Acetylcholine is the main neurotransmitter of the REST AND DIGEST nervous system, so anticholinergic agents decrease GIT actions.

14

Anticholinergic agents

(against acethycholine)

Hyoscine (scopolamine)

Acetylcholine is the main neurotransmitter of the REST AND DIGEST nervous system, so anticholinergic agents decrease GIT actions.

* Indicated in spasms and/or motion sickness
* Main side effect: dry mouth, constipation

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Antihistamines that break the blood brain barrier (Diphenhydramine and dimenhydrinate)

Histamine in the brain is important in maintaining arousal

Side effects: drowsy, decreased awareness

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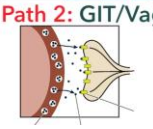
Vomiting centre

*Uses H1 and ACh

*Activated by 4 paths

Path 3: Chemoreceptor trigger zone (CRTZ)

Path 2: GIT/Vagal nerve



Both paths use serotonin (5HT3),

*Selective 5HT3 ANTAGONISTS (...setron drugs) are effective in decreasing these impulses

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The next few slides picture the drugs that help with nausea from chemotherapy and GIT irritation

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* -SETRON drugs are 5HT receptor blockers
* One of the most effective antiemetic for chemotherapy induced vomiting
* Relatively few side effects

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ProchlorperazIne, metoclopramide and other dopamine 2 antagonists

* Fairly effective for emesis originating in the GIT or the CRTZ
* Inexpensive
* Many side effects including movement disorders

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The next few slides picture the drugs that are good for each type of nausea

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Diphenhydramine
A histamine 1 antagonist



* Relieves motion sickness (and inner ear vomiting)
* No relief is expected in chemotherapy vomiting

22

Hyoscine (Scopolamine)
An anticholinergic agent



* Relieves motion sickness
* May relieve spasms and GIT overactivity

23

Lorazepam
A benzodiazepine



* Treats emesis originating in the cerebellum
* Occasionally used in combination with several other medications for chemotherapy emesis

24

Metoclopramide
A dopamine 2 antagonist



* May be effective in high doses
* May cause movement disorders and several other side effects

25

Ondansetron
A serotonin 3 antagonist



* The most effective medication for chemotherapy-induced nausea and vomiting
* Expensive

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References

- Dr Ken Wojcikowski
- PharmacoPhoto
- Royalty free images provided by Shutterstock

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